Chuẩn bị dữ liệu và HDFS

```
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ echo "Hadoop is fast. Hadoop is powerful. YARN is a resource negotiator. UET student s
tudy Big Data." > uet_data.txt
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker cp uet_data.txt namenode:/tmp/uet_data.txt
Successfully copied 2.05kB to namenode:/tmp/uet_data.txt
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker exec -it namenode hdfs dfs -mkdir -p /user/vanvan/input
docker exec -it namenode hdfs dfs -put /tmp/uet_data.txt /user/vanvan/input
2026-03-13 03:49:47,340 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker exec -it namenode hdfs dfs -ls /user/vanvan/input
Found 1 items
-rw-r--r--   3 root supergroup         95 2026-03-13 03:49 /user/vanvan/input/uet_data.txt
```

Chạy Job MapReduce

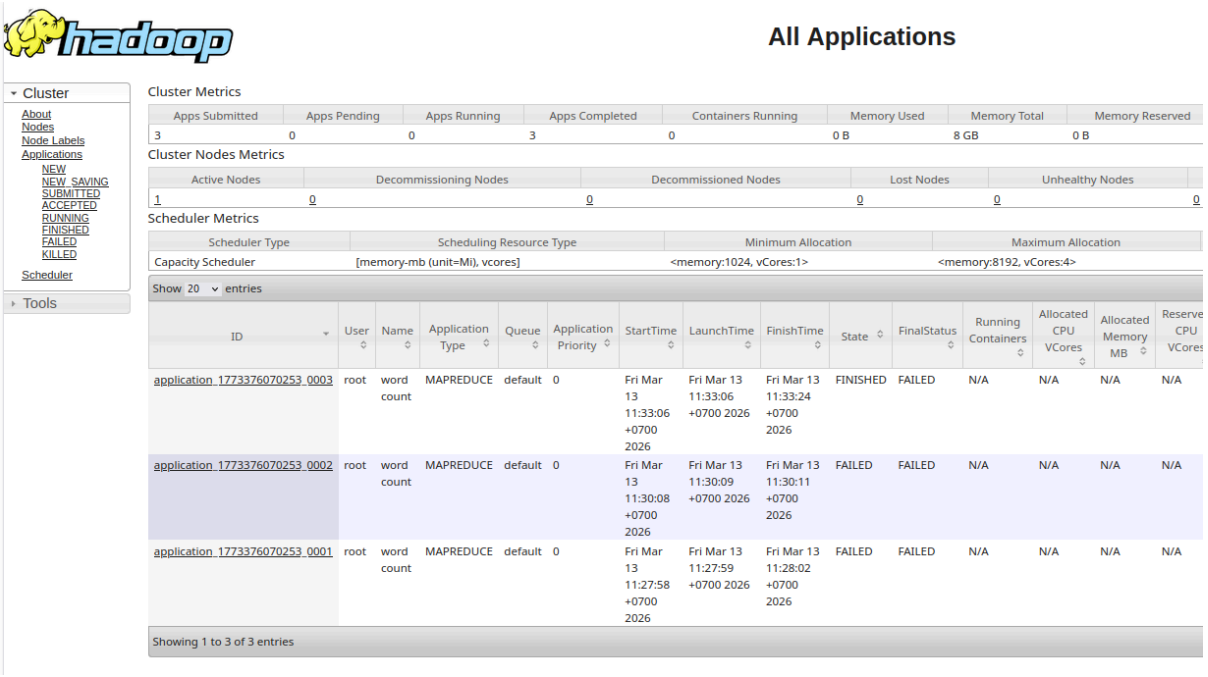
```
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker exec -it namenode hadoop jar /opt/hadoop-3.2.1/share/hadoop/mapreduce/hadoop-ma
preduce-examples-3.2.1.jar wordcount /user/vanvan/input /user/vanvan/output
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=13
                Reduce shuffle bytes=166
                Reduce input records=13
                Reduce output records=13
                Spilled Records=13
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=370147328
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=108
2026-03-13 03:52:20,168 INFO mapred.LocalJobRunner: Finishing task: attempt_local725300431_0001_r_000000_0
2026-03-13 03:52:20,168 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-13 03:52:21,166 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-13 03:52:21,167 INFO mapreduce.Job: Job job_local725300431_0001 completed successfully
2026-03-13 03:52:21,179 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=633838
                FILE: Number of bytes written=1679548
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=190
```

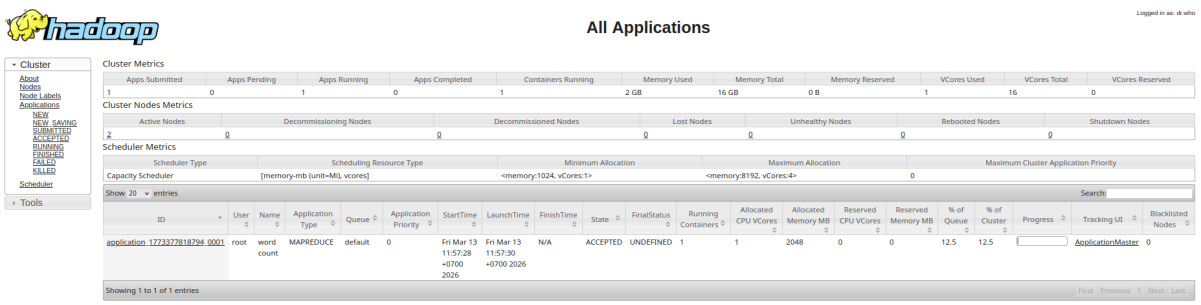
kiểm tra kết quả

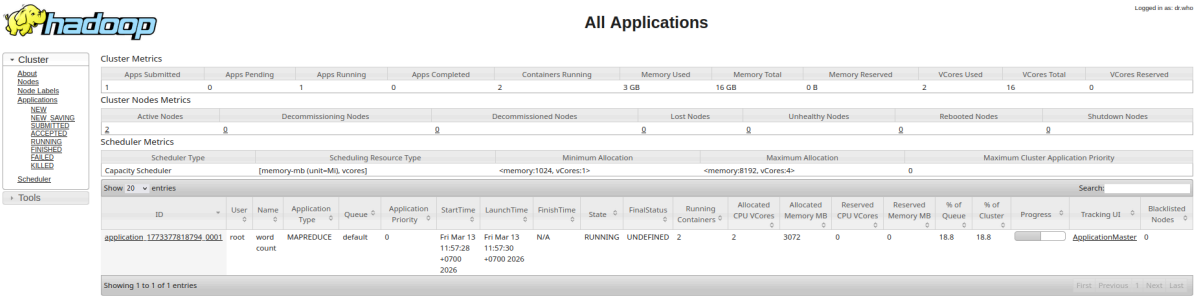
```
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker exec -it namenode hdfs dfs -cat /user/vanvan/output/part-r-00000
2026-03-13 03:54:46,106 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
Big     1
Data.   1
Hadoop  2
UET     1
YARN    1
a       1
fast.   1
is      3
negotiator.     1
powerful.       1
resource        1
student 1
study   1
```

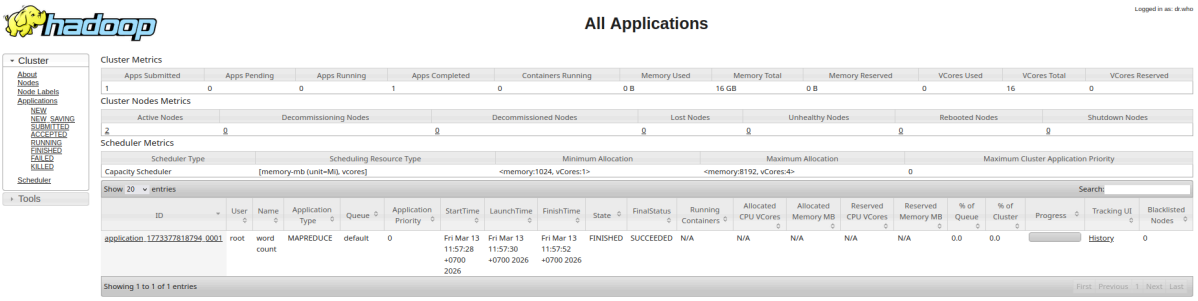
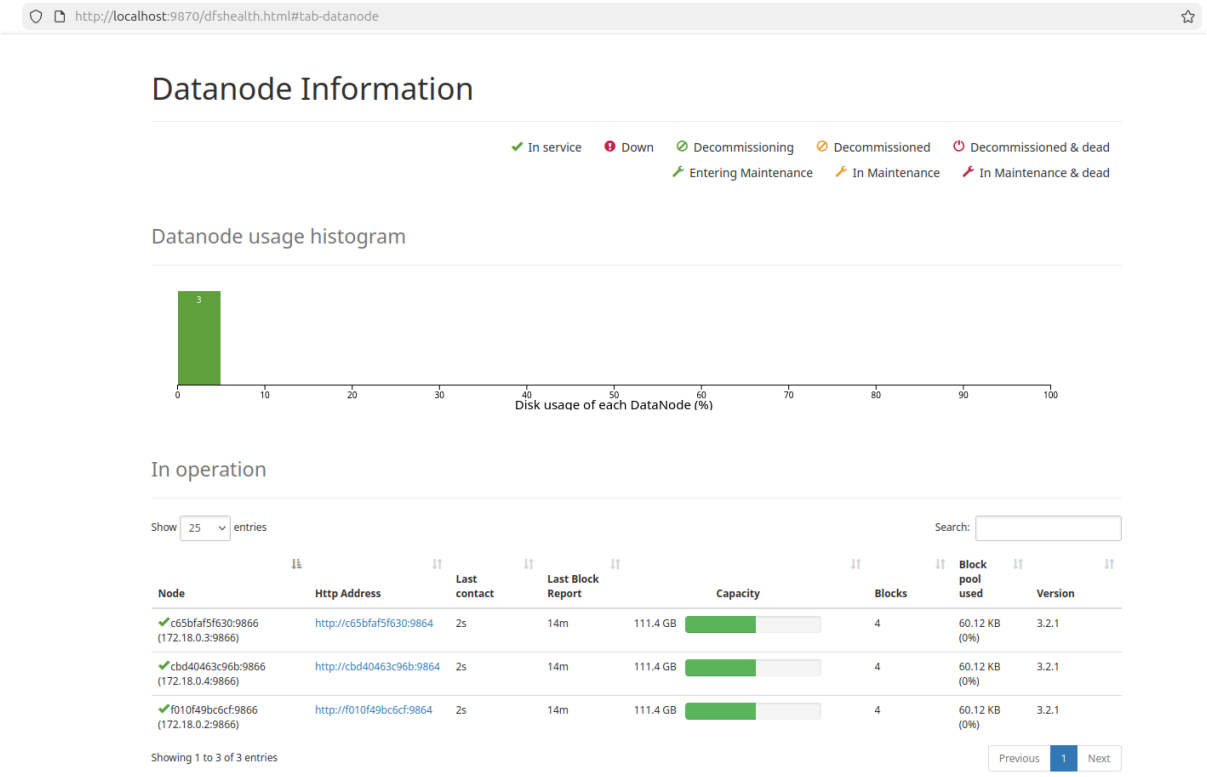
```
# Chạy Job (Lúc này nhìn vào localhost:8088 sẽ thấy nó hiện lên ngay!) …
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=340095
        File Output Format Counters
                Bytes Written=150
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$ docker exec -i
t namenode hdfs dfs -cat /user/vanvan/output/part-r-00000
2026-03-13 05:00:10,150 INFO sasl.SaslDataTransferClient: SASL encryption trust check
 localHostTrusted = false, remoteHostTrusted = false
Big     1
BigData 10000
Data.   1
Hadoop  10002
MapReduce       10000
UET     10001
YARN    10001
a       1
fast.   1
is      3
negotiator.     1
powerful.       1
resource        1
student 1
study   1
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Năm 3/BigData$
```

YARN









Trạng thái các node